$\beta_0 = 1,$

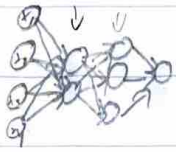
For 1: let $\beta_1 = 1$, $\beta_2 = 2$, etc.

Jan 14:
Book #1 & 2
Supp #1, 2, 6, 7
R book #7

let $z_+(x) = \text{ReLU for } x = \begin{cases} 0 \text{ if } x < 0, \\ x \text{ otherwise} \end{cases}$

layer 1 layer 2

1 a)



b) $f(x) = \sum_{m=1}^{3} w_m + \sum_{l=1}^{3} w_l \left(\beta_0 + \sum_{k=1}^{2} \beta_k z_+(w_{k0} + \sum_{j=1} w_{kj} X_j)\right)$

d) $5 \times 2 = 10 +$
$3 \times 3 = 9 +$
$4 \times 1 = 4 = 23$ weights

For $X \to$ first layer, then plug all that in for first $\to$ second layer

c) $f(x) = 8 + 5(z_+(w_5 + [w_1(1 + 1 z_+(w_{1,0} + w_{1,1}X_1 + w_{1,2}X_2 + w_{1,3}X_3 + w_{1,4}X_4) +$
$2 z_+(w_{2,0} + w_{2,1}X_1 + w_{2,2}X_2 + w_{2,3}X_3 + w_{2,4}X_4)) +$
$w_2(1 + 1 z_+(w_{1,0} + w_{1,1}X_1 + w_{1,2}X_2 + w_{1,3}X_3 + w_{1,4}X_4) +$
$2 z_+(w_{2,0} + w_{2,1}X_1 + w_{2,2}X_2 + w_{2,3}X_3 + w_{2,4}X_4)) +$
w_3
$]))+$

$6(z_+(w_6 + [w_1(1 + \ldots)]))$
$7(z_+(w_6 + [w_1(1 + \ldots)]))$

2 a) $\text{softmax}_m(x) = \frac{e^{z_m}}{\sum_{l=m}^{M} e^{z_l}}$ $\sum e^{z_l} = e^{z_1} + e^{z_2}$

w/ constant $= e^{c_1} + e^{c_2} + \ldots + e^{c_m} = me^{c}$ $= me^{c}(e^{z_1} + e^{z_2} + \ldots + e^{z_m})$

by doing this in the numerator also, the me^{c} cancel, giving us the same probability.

b) Num: $e^{(c_0 + \beta_{k0}) + (c_1 + \beta_{k1})x_1 + (c_2 + \beta_{k2})x_2 + \ldots}$
Denom: $\sum_{l=1}^{k} e^{(c_0 + \beta_{l0}) + (c_1 + \beta_{l1}x_1) + (c_2 + \beta_{l2}x_2) + \ldots}$

By adding constants, again both the denom & num can factor out & cancel $e^{c_0} \cdot e^{c_1 x_1} \cdot e^{c_2 x_2} \ldots$ by splitting up the exponents with e nicely.

$= 160(\hat{y}_i - 1) + 40(\hat{y}_i)$

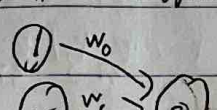
180 times, 20 times

Supp 1 Error: $\sum_{i=1}^{100}(y_i - f(x_i))^2$ $\quad \frac{d}{dx}\sum_{i=1}^{100}(\hat{y}_i - f(\hat{x}_i))^2 = \sum_{i=1}^{80} 2(y_i - 1) + \sum_{i=1}^{20} 2(y_i - 0)$

optimum is 8, since $\frac{dy}{dx}$ of loss = 8, and switches from − to +, so this is an inflection point of the loss graph, and is the global minimum

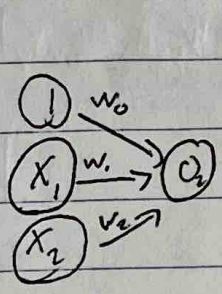
Supp 2 A network of activation functions that are linear are very uninteresting because a linear combination of a linear combination results in a linear combination. With an output of this network, it is simply a linear combination of linear combinations of the inputs. In this case, we should just train a regular regression model to find the "weights" or coefficients to this linear combination of the inputs

Supp 6

a) 

$w_0 = 1$ $\quad$ If $x = 0$ (False), $O_1 = 1 + 0 \cdot -1 = 1$ (TRUE)

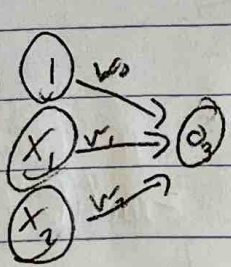
$w_1 = -1$ $\quad$ If $x = 1$ (TRUE), $O_1 = 1 + 1 \cdot -1 = 0$ (FALSE)

b) 

$W_0 = 0$ $\quad$ If both X_1 & $X_2 = 0$, then out is 0. F

$W_1 = 1$ $\quad$ either or both X_1 & $X_2 = 1$, then out is 0 = TRUE

$W_2 = 1$

c) 

$W_0 = -1$ $\quad$ Both X_1 & X_2 have to be TRUE for $O_3 = 1$, which is TRUE

$W_1 = 1$

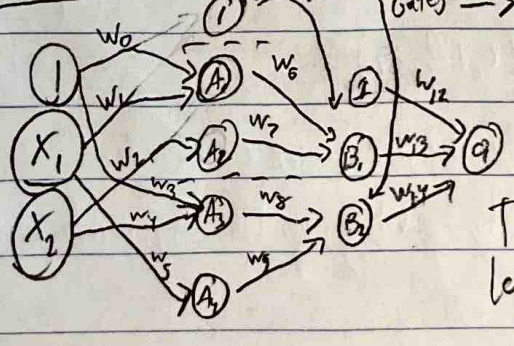
$W_2 = 1$

All of this can be two NOT Gates, $\rightarrow$ $W_0 = 1, w_1 = -1, w_2 = 1, w_3 = 1, w_4 = -1, w_5 = 1$

2 AND Gates $\rightarrow$ $W_{10} = W_{11} = -1$, $W_6 = W_7 = W_8 = W_9 = 1$

$W_{12} = 0$, $W_{13} = W_{14} = 1$

Supp 7 XOR:



This is the last OR Gate!